DBMS Lab: Transaction Control

Topic: BEGIN, SAVEPOINT and ROLLBACK

Scenario: Inventory Restocking with Savepoint

A warehouse manager restocks the product with **product_id = 5** in two batches of **10 units** each. After adding the first batch, he creates a savepoint named **s1**. He then adds the second batch, but realizes it was entered by mistake. He must undo only the second batch, keep the first batch, and save the change permanently.

Database Table

product (product_id *[PK]*, name, stock)

Initial data:

product_id	name	stock
5	Keyboard	50

Part A: Single Savepoint

1. Create the **product** table and insert the initial data.
2. Start a transaction using **BEGIN**.
3. Add the first batch of 10 units to product_id 5.
4. Create a savepoint named **s1**.
5. Add the second batch of 10 units to product_id 5.
6. Undo only the second batch using **ROLLBACK TO SAVEPOINT**.
7. Save the transaction permanently using **COMMIT**.
8. Display the final row of product_id 5.

Part B: Multiple Savepoints

Continue with the stock value saved at the end of Part A. In a new transaction, the manager records several changes for product_id 5 and creates a savepoint after each one. Some of these changes later turn out to be wrong.

9. Start a new transaction using **BEGIN**.
10. Add **5** units (small delivery). Create savepoint **s1**.
11. Add **15** units (large delivery). Create savepoint **s2**.
12. Subtract **8** units (sold to a customer). Create savepoint **s3**.
13. Add **20** units (entered by mistake).
14. Undo only the wrong 20-unit entry using **ROLLBACK TO SAVEPOINT s3**. Display the stock.
15. The sale of 8 units was cancelled. Undo it using **ROLLBACK TO SAVEPOINT s2**. Display the stock.
16. The large delivery of 15 units was also wrong. Undo it using **ROLLBACK TO SAVEPOINT s1**. Display the stock.
17. Save the transaction using **COMMIT** and display the final row of product_id 5.